Emerging Technologies for the Circular Economy

Lecture 6: Internet of Things Communications

Prof. Dr. Benjamin Leiding
M.Sc. Anant Sujatanagarjuna
M.Sc. Chintan Patel

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NEWS/UPDATES

Course Evaluation – QR Code and Link

- **For the course: „*Emerging Technolgoies for the Circular Economy*"**
- Link: Click Me

Course Evaluation – QR Code and Link

- ***For the course: „IoT and Digitalization for Circular Economy"***
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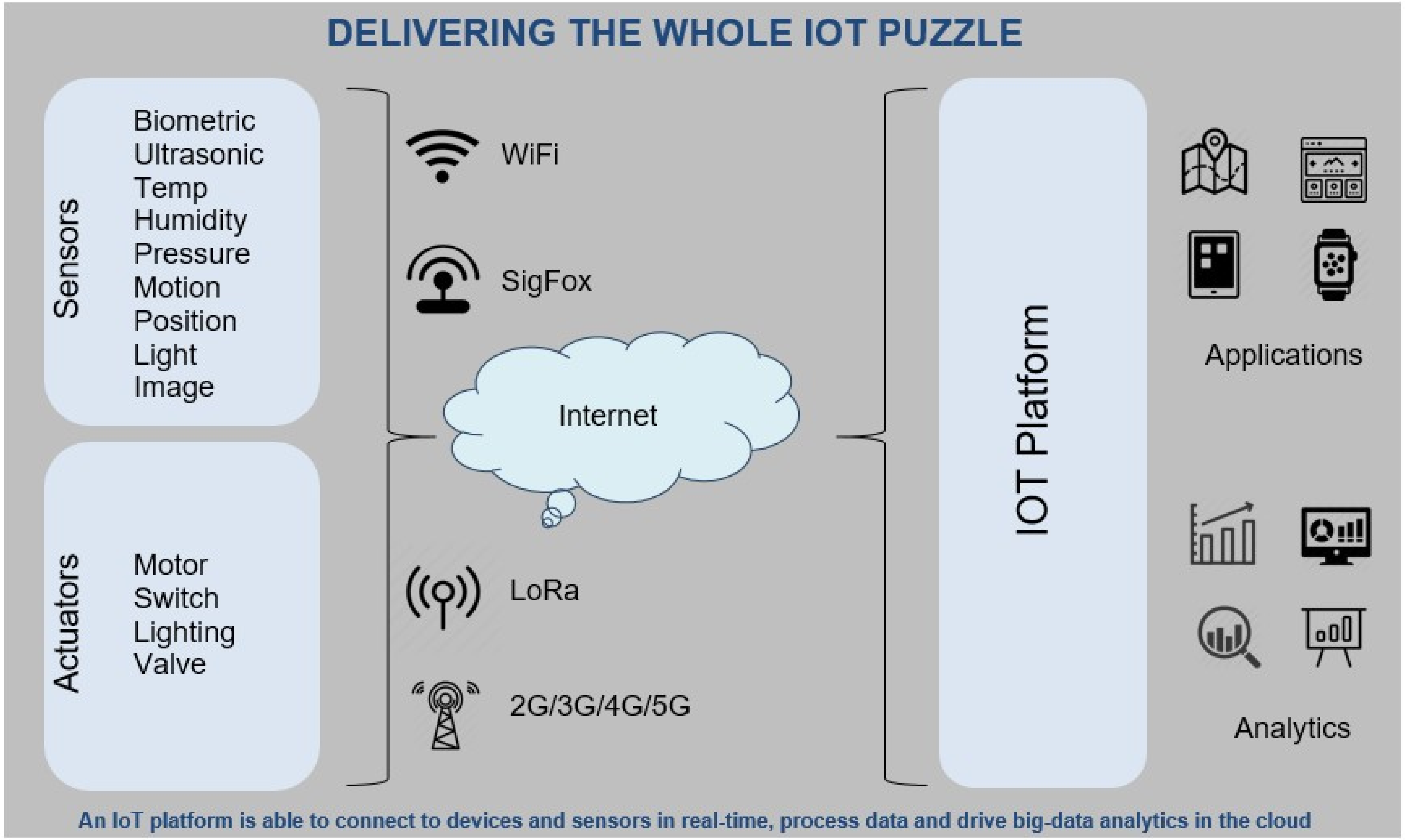
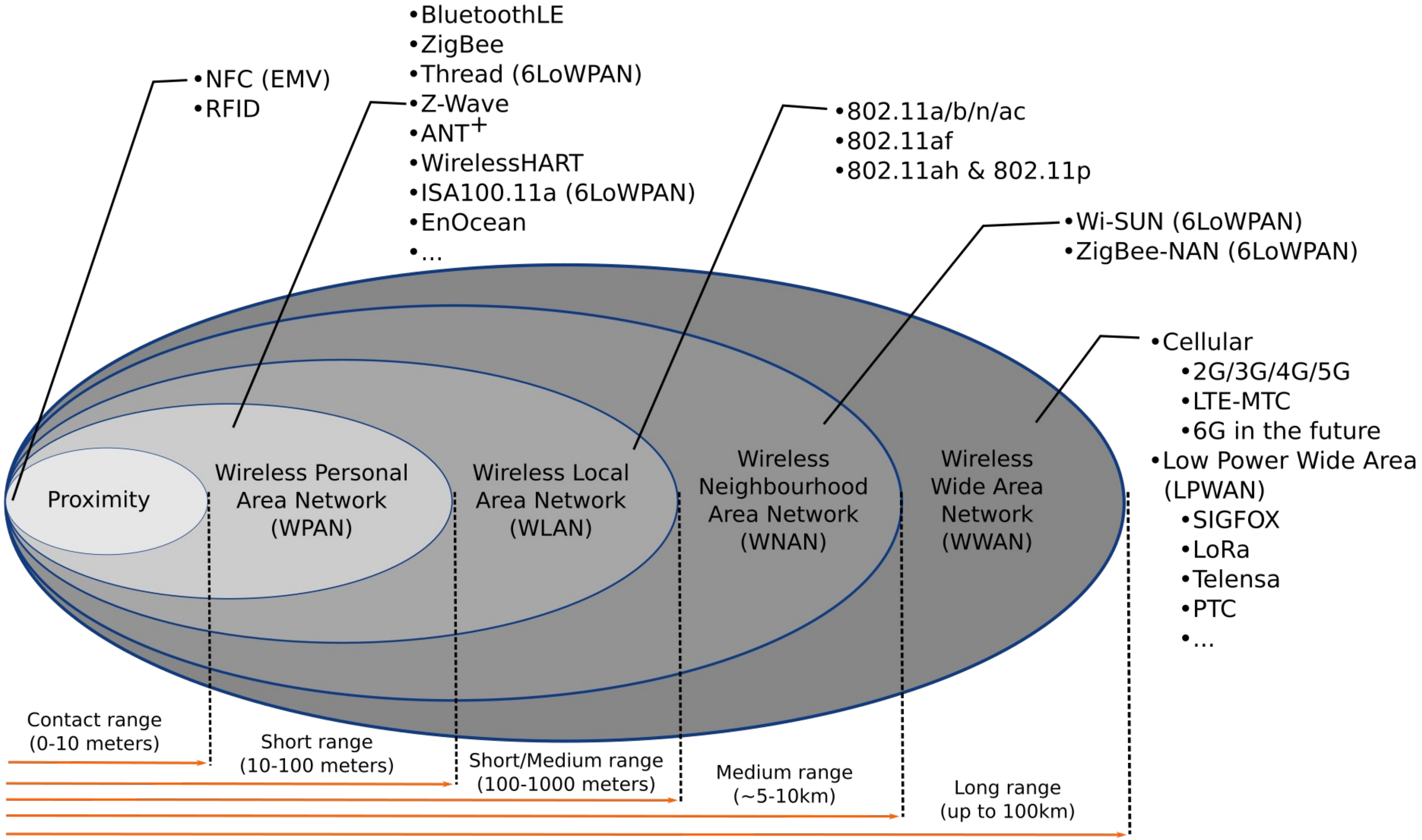
IOT COMMUNICATIONS

Overview


DELIVERING THE WHOLE IOT PUZZLE
Sensors
Biometric
Ultrasonic
Temp
Humidity
Pressure
Motion
Position
Light
Image
Actuators
Motor
Switch
Lighting
Valve
WiFi
SigFox
Internet
LoRa
2G/3G/4G/5G
IOT Platform
Applications
Analytics
An IoT platform is able to connect to devices and sensors in real-time, process data and drive big-data analytics in the cloud

Different ranges, different standards



Source: M.S. Mahmoud, A. Mohamad, "A Study of Efficient Power Consumption Wireless Communication Techniques/ Modules for Internet of Things (IoT) Applications", January 2016, Advances in Internet of Things 06(02):19-29, DOI:10.4236/ait.2016.62002

WPAN

Wireless Personal Area Network

IP-Based

Protocol	Description	Power Consumption	Usage
6LoWPAN	IPv6 over Low-Power Wireless Personal Area Networks	Low (< 100 mW)	IPv6-based IoT networks
IEEE 802.11p (V2V)	WiFi-based	High (>1W)	Vehicle-to-vehicle (V2V) communications
RuBee	Uses magnetic fields over short range	Very Low (<10 mW)	Harsh environments / tracking high-value assets

Wireless Personal Area Network

Not IP-Based

Protocol	Description	Power Consumption	Usage
Bluetooth/ BLE	Versatile and high data throughput per watt	Low (< 100 mW)	Varies by specific protocol
ZigBee	Low-power, Low data-rate	Low (< 100 mW)	Vehicle-to-vehicle (V2V) communications
IrDA	Infrared Data Association (Only works with line-of-sight)	Medium (100-500 mW) over short bursts	Remote controls
Z-Wave	Similar to ZigBee and BLE, but also supports mesh networking	Low (25-80 mW)	Home automation, sensors, locks, etc.

Bluetooth

Pros
- Low power requirements
- Resilient against interference

Cons
- Low bandwidth
- Limited range
- Limited number of participants in network

Applications
- Beacons
- Fitness trackers, smart watches
- Medical applications
- Smart homes
- Smart cars
- Earbuds, headsets etc.

Classes for different applications with different ranges/power usages.

Zigbee

Pros
- Low power requirements
- Scales to large network sizes (~6500 nodes)

Cons
- Low range
- Low bandwidth
- Security issues (fixed, known fallback keys in at least one profile)

Applications
- Wireless sensor networks (WSN)
- Industrial automation
- Smart homes

6LoWPAN

Pros
- IPv6 based
- Built-in security
- Scalability
- Interoperability

Cons
- Higher minimum requirements due to IPv6 minimum complexity
- Not as popular as ZigBee

Applications
- Wireless sensor networks (WSN)
- Internet of Things
- Industrial Internet of Things

IEEE 802.11p

Vehicular network optimized
- Vehicle to vehicle (V2V)
- Vehicle to infrastructure (V2I) such as road side units (RSU)
- Built in time synchronization

Applications
- Vehicular networks

WAN

Wide Area Network

- Service/subscription model based
- Service provider runs infrastructure such as base stations and radio towers

Protocol	Power Consumption	Data throughput
2G (GSM)	~1-2 W	0.03-0.24 Mb/s
3G (UMTS/HSPA)	~1-1.5 W	1-5 Mb/s
4G (LTE 1/4/6)	~400 mW – 1.5W	5-150 Mb/s
5G	~500 mW – 2W+	50 Mb/s – 1+ Gb/s
LoRaWAN	~10 mW – 100 mW	0.3 – 50 kb/s
SigFox	~10 mW – 80 mW	0.1 – 0.6kb/s

Cellular network architecture

- Grid of cell towers
- Overlapping cells
- Requires handover for mobile stations between cells

- Network planning
- Space division multiple access
- Minimize interference
- Avoid allocating overlapping spectrum on nearby cells

5G

- New radio communication techniques and spectrum
- Support for device to device communications (D2D)
- Improved performance
  - Theoretical latency in single digit ms
  - Bandwidth in gbps range
  - Ability to provide connectivity in fast moving vehicles
  - Enables more dense connectivity and scalability (more devices)

LoRa/LoRaWAN

- Uses unlicensed spectrum
- Low number of base stations (Gateways) covers wide area
  - 7 are enough to cover Belgium
- Only produced by a single company (Simtech)
- High latency, no realtime applications
- Subscription based
- Misses some common features from LTE networks
  - Only physical and MAC layers are covered => Higher OSI layers have to be implemented on top

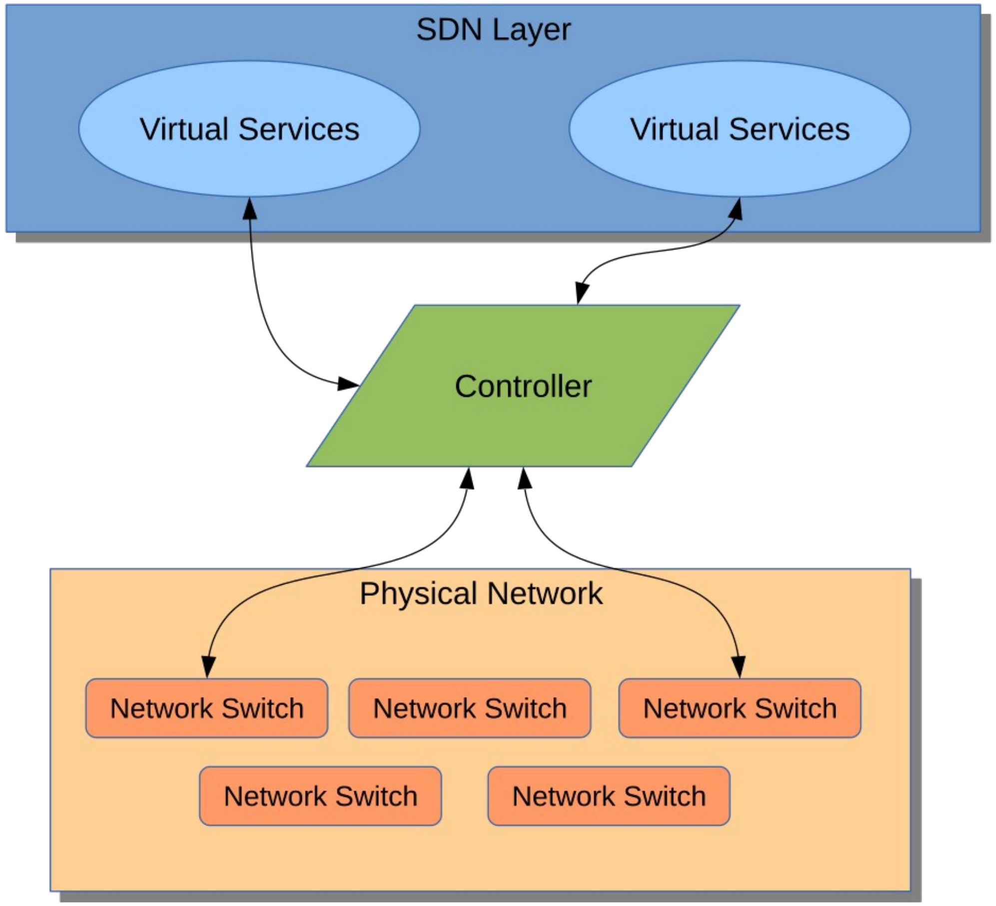
Sigfox

- Uses unlicensed spectrum
- Uplink
  - 100bps
  - 12B payloads
  - Maximum of 6 messages per device and hour (140 per day)
- Downlink
  - 600bps
  - 8B payloads
  - Maximum of 4 messages per day
- Open hardware
- Network subscription based

SOFTWARE-DEFINED NETWORKS (SDN)

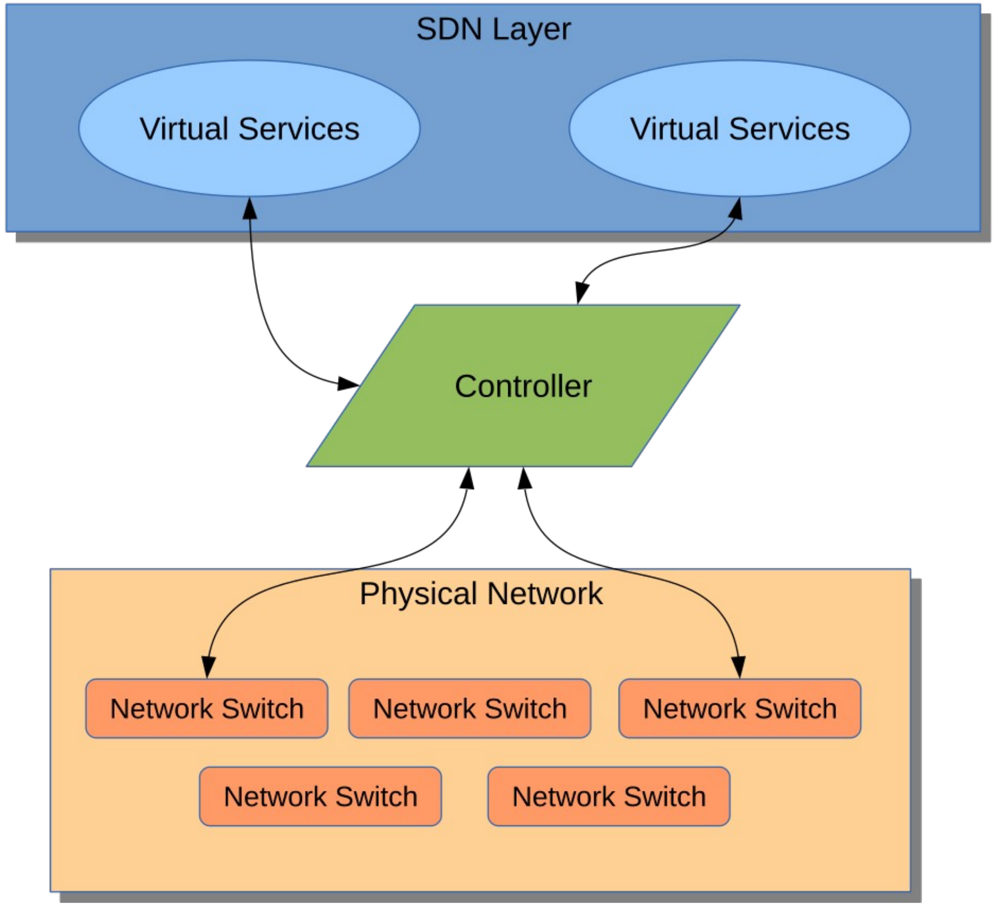
SDN Architecture

- SDN: Umbrella term used for when the physical networking layer is abstracted from the logical layer.
- All physical networking devices simply forward data packets to a centralized controller.
- Virtual services (apps/databases/IoT applications) also rely on communication with the central controller to interface with other IoT components like sensors, actuators, etc.
- Only the controller is “aware” of the physical networking architecture in relation to the virtual services and routes network traffic accordingly.



SDN Architecture

- Salient Features:
  - Programmatically more flexible: network managers can optimize network resources dynamically and automatically.
  - The controller is generally centralized – simplifying implementation of control logic.
  - Centralization might introduce more issues such as security, scalability and reliability.
  - Open protocols and standards for SDN (like OpenFlow) can be used on top of proprietary networks for added privacy.
  - Especially suited to cloud-based IoT systems.
  - Significantly higher energy usage!



Khan, Sahrish & Shah, Munam & Khan, Omair & Ahmed, Abdul. (2017). Software Defined Network (SDN) Based Internet of Things (IoT): A Road Ahead. 1-8. 10.1145/3102304.3102319.

ROUTERS AND GATEWAYS

Routers and Gateways

Router

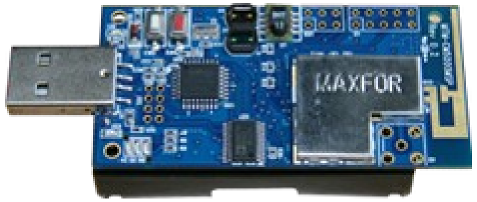
- Bridges two networks
- Can translate between protocols
- Routes data
- Port forwarding and network address translation (mainly end user or carrier grade)
- VNETs

Gateway (not in the routing sense)

- Bridges wireless network and internet
- Can translate between protocols
- Edge/Fog computing capabilities (Lecture 8: IoT Data processing and Big Data)
- Routers can be gateways

Gateway example

- Wireless sensor nodes running Contiki RPL with Ipv6
- Node attached to gateway over USB acts as gateway
- IPv6 connectivity between networks provided through SLIP (Serial Line Internet Protocol)



Questions?